

Erdős Problem #124

EP-124

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For any $d \geq 1$ and $k \geq 0$ let $P(d, k)$ be the set of integers which are the sum of distinct powers d^i with $i \geq k$. Let $3 \leq d_1 < d_2 < \dots < d_r$ be integers such that $\sum_{1 \leq i \leq r} \frac{1}{d_i - 1} \geq 1$. Can all sufficiently large integers be written as a sum of the shape $\sum_i c_i a_i$ where $c_i \in \{0, 1\}$ and $a_i \in P(d_i, 0)$?

If we further have $\gcd(d_1, \dots, d_r) = 1$ then, for any $k \geq 1$, can all sufficiently large integers be written as a sum of the shape $\sum_i c_i a_i$ where $c_i \in \{0, 1\}$ and $a_i \in P(d_i, k)$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>